

MACHA, Russian Federation

WMO: 300740

Lat: 59.904N

Lon: 117.588E

Elev: 537

StdP: 14.41

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-52.4	-47.7	-58.8	0.2	-42.5	-52.6	0.2	-47.5	20.2	5.7	14.4	4.8	0.6	330	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	84.7	65.4	80.2	63.9	76.3	62.1	68.1	80.2	66.1	76.5	64.1	73.1	6.4	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.1	91.5	72.6	62.1	85.1	70.3	60.3	79.8	68.1	32.8	80.5	31.2	76.4	29.7	73.4	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.6	14.6	12.3	DB	-53.6	91.1	3.7	3.9	-56.3	93.8	-58.4	96.1	-60.5	98.3	-63.2	101.1	(4)
(5)			WB	-54.2	71.3	6.3	2.6	-58.7	73.2	-62.3	74.8	-65.8	76.2	-70.4	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.9	-21.8	-11.4	9.8	29.6	44.4	59.0	63.5	59.5	44.7	26.5	-2.3	-17.6
	DBStd	32.42	16.72	14.64	14.56	9.74	8.35	6.73	6.20	6.43	7.13	10.14	15.73	17.67
	HDD50	10611	2225	1720	1245	611	214	7	0	8	190	728	1568	2095
	HDD65	15116	2690	2140	1710	1061	638	200	103	194	609	1193	2018	2560
	CDD50	1072	0	0	0	0	42	279	419	301	31	0	0	0
	CDD65	101	0	0	0	0	0	22	57	22	0	0	0	0
	CDH74	1292	0	0	0	0	43	348	657	241	3	0	0	0
	CDH80	375	0	0	0	0	9	84	217	65	0	0	0	0
Wind	WSAvg	4.3	2.8	3.4	4.7	5.9	5.7	4.9	4.3	3.8	4.5	5.1	3.7	3.0
Precipitation	PrecAvg	18.9	0.7	0.6	0.5	0.9	1.8	2.5	3.5	3.0	2.2	1.3	1.1	0.8
	PrecMax	24.6	1.1	1.3	0.9	2.2	3.6	4.2	6.0	6.1	3.9	2.1	1.7	1.4
	PrecMin	12.6	0.3	0.2	0.1	0.4	1.0	1.4	1.4	0.9	1.0	0.7	0.3	0.3
	PrecStd	2.7	0.2	0.2	0.2	0.4	0.7	0.7	1.4	0.9	0.7	0.4	0.4	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	25.6	47.8	57.1	79.3	87.8	91.4	87.3	70.7	52.7	43.0	28.2
		MCWB	14.8	21.8	38.1	44.1	57.8	63.9	68.1	66.0	57.2	43.1	37.9	24.7
	2%	DB	13.1	20.8	40.0	51.9	71.5	82.9	86.0	80.3	64.3	47.6	30.5	23.0
		MCWB	11.3	18.3	32.8	40.1	53.4	62.5	66.7	65.2	53.8	41.1	27.4	21.0
	5%	DB	8.5	16.4	35.0	48.6	66.1	78.7	81.3	75.9	59.7	43.3	24.3	17.2
		MCWB	7.1	14.5	29.2	38.1	50.0	61.4	66.0	62.8	51.7	37.2	21.5	15.7
	10%	DB	2.8	9.9	30.4	44.6	60.3	74.6	76.7	72.2	56.0	39.6	19.5	9.5
		MCWB	1.9	8.3	26.2	36.0	47.1	59.4	63.8	61.2	49.2	34.9	17.9	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	21.9	38.0	44.9	58.5	68.3	71.6	69.6	58.4	44.3	37.8	25.2
		MCDB	17.1	24.9	47.2	54.5	79.4	80.9	83.7	79.7	69.0	50.3	43.7	27.6
	2%	WB	11.5	17.9	33.2	41.2	54.1	65.5	69.2	67.1	55.5	41.4	27.5	21.0
		MCDB	13.1	20.2	39.7	50.9	69.9	77.9	82.7	76.7	62.0	46.9	30.8	23.4
	5%	WB	7.0	14.1	29.5	38.8	51.0	63.3	67.1	65.0	52.9	38.1	21.6	15.9
		MCDB	8.4	16.2	34.3	47.0	63.9	74.8	78.7	73.2	58.1	42.8	23.9	17.4
	10%	WB	1.9	7.9	26.4	36.5	48.1	61.0	65.0	63.0	50.2	35.3	17.7	9.0
		MCDB	2.9	9.3	30.6	44.1	58.7	71.8	74.7	70.5	55.2	39.6	19.6	10.0
Mean Daily Temperature Range	5% DB	MDBR	14.8	18.7	23.5	20.0	19.8	21.8	19.6	19.6	16.4	12.0	14.0	13.4
		MCDBR	18.8	21.7	23.2	24.1	31.8	30.6	29.3	29.1	24.6	16.2	18.1	16.2
		MCWBR	17.2	20.6	16.7	14.2	15.8	13.3	12.5	14.2	13.8	11.0	15.6	14.6
	5% WB	MCDBR	18.5	21.9	21.7	21.0	29.5	23.5	23.9	22.3	21.2	15.1	17.4	16.7
		MCWBR	17.0	20.0	15.8	12.8	15.4	11.5	11.4	11.4	12.7	10.5	15.2	15.2
Clear-Sky Solar Irradiance	taub	0.250	0.284	0.311	0.355	0.393	0.434	0.481	0.427	0.340	0.297	0.253	0.224	
	taud	1.976	2.027	2.062	2.124	2.252	2.230	2.094	2.229	2.448	2.356	2.036	1.921	
	Ebn at Noon	156	220	255	265	264	253	235	240	250	223	154	117	
	Edh at Noon	16	26	35	40	38	40	45	36	25	19	14	11	
All-Sky Solar Radiation	RadAvg	148	436	942	1493	1599	1751	1588	1253	783	385	187	82	
	RadStd	10	29	60	89	105	166	165	75	76	34	22	7	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page